relationships between states in quantum mechanics are always linear. It goes along with the idea that the state-space is a vector space. But linearity is not the only thing that quantum mechanics requires of **U**(*t*). It also requires the quantum analog of the minus first law: *the conservation of distinctions*.

Recall from the last lecture that two states are distinguishable if they are orthogonal. Being orthogonal, two different basis vectors represent two distinguishable states. Suppose that |Ψ(0)> and |ϕ(0)> are two distinguishable states; in other words, there is a precise experiment that can tell them apart, and therefore they must be orthogonal:

$$\langle \Psi(0) | \Phi(0) \rangle = 0.$$

The conservation of distinctions implies that they will continue to be orthogonal for all time. We can express this as

$$\langle \Psi(t) | \Phi(t) \rangle = 0 \qquad (4.2)$$

for all values of *t*. This principle has consequences for the time-development operator **U**(*t*). To see what they are, let's flip the ket-vector Eq. 4.1 to its bra-vector counterpart:

$$\langle \Psi(t) | = \langle \Psi(0) | \mathbf{U}^\dagger(t). \qquad (4.3)$$

Notice the dagger that indicates Hermitian conjugation. Now, let's plug Eqs. 4.1 and 4.3 into Eq. 4.2:

$$\langle \Psi(0) | \mathbf{U}^\dagger(t) \mathbf{U}(t) | \Phi(0) \rangle = 0. \qquad (4.4)$$

To examine the consequences of this equation, consider an orthonormal basis of vectors |*i*>. Any basis will do. The orthonormality is expressed in equation form as

$$\langle i | j \rangle = \delta_{ij},$$